

Question justifications

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1 Pre-session Questionnaire

1.1 Participant Experience Information

1. What is your name?
2. How old are you?
3. What gender do you identify with?
4. What is your occupation?

Questions 1 – 4: These are used to confirm the identity of the participant being interviewed.

5. Have you ever had a knee injury? **Yes/No/NA**
6. If **YES**, was it in the past 5 years?
7. If **YES**, have you had physiotherapy after your knee surgery?
8. If **NO**, have you had another kind of orthopaedic surgery?
9. If **NA**, do you personally know anyone who has experienced a knee injury?
YES/NO
10. How long did your post-operative physiotherapy last?
11. Were you prescribed knee extensions as an exercise to do at home?
12. In your opinion, did you encounter any trouble performing knee extensions using the application? **YES/NO**

This examines the participant's experiences. In an ideal case, the participant will have personally had a knee injury in the past. '5 years' is a conservative amount of time that was chosen, because technology is ever-changing, and is projected to change in the near future (<https://www.hunimed.eu/news/10-ways-technology-is-changing-healthcare/>). In addition, the level of care that was received previously may have been very different than what is practiced today.

Experience with a personal orthopaedic surgery has a minor overlap with having a knee surgery due to their similar nature. In that sense, an orthopaedic patient may be more empathetic to those that have had surgeries recently. They may also impart their first-hand understanding of such an experience and provide richer, more nuanced insights into the potential benefit and use of the application.

1.2 Participant device literacy

9. What do you use your mobile device for?
10. How comfortable are you using the camera function?
Very comfortable Somewhat comfortable Comfortable
Not comfortable
11. Do you know what a mobile phone tripod is? (selfie stick)
12. How comfortable are you using the tripod?
Very comfortable Somewhat comfortable Comfortable
Not comfortable

In order to properly operate the application, there is a minimum level of device literacy needed. These questions specifically target smartphone users. Some populations may not have equal access to the level of care (socioeconomic reasons) that is assumed for this application's use, and that is somewhat portrayed through the use of these questions. The purpose of these questions asserts the need for further exploration if/when the target audience is widened and should further development of the application occurs.

1.3 Participant perspective on privacy

1. Participant identification information will be stored on a database.
2. Information extracted from the recording used in the application will also be stored on a database.
3. The database used in the application adheres to data privacy standards and regulations.
4. All participant data will be deleted within (2 weeks?) after storage.
5. Information access is restricted to any external parties (except the application developers for the duration mentioned above).
6. All participant data will be deleted within (2 weeks?) after storage.

This section outlines some private aspects involved in using the app. These statements should be confirmed with participants to ensure that a well-informed decision to participate and use it has been made.

2 Session questionnaire

The account creation functionality is a similar one to that of existing apps. It uses a conventional flow that should be familiar to users. The ease of this process is evaluated. Once a participant has created an account, they are directed

to the Homescreen. This page has different capabilities based on the type of user.

On a patient's screen, it displays a history of completed exercises and their dates. On a PT's screen, it displays the names of their patients. Once they have selected a patient, the screen is populated with the patient's activity. Both the patient's and PT's homescreen only serve as a record, displaying patient activity history. Since both patient and PTs pages look very similar, the main questions posed for this section are UI perspectives. Drawing on both perspectives, these questions serve to validate the homescreen's minimal aesthetic, confirming that it contains only essential information and is not visually cluttered. The same logic follows the progress tab.

Despite the UI being almost the same, there is a functional difference when the two types of users interact with the Exercises page which is addressed by the questions.

In the case of the patient, the questions aim to determine whether the elements are presented clearly and in a straightforward manner. The example video under the complete exercise instructions was implemented at a later stage, necessitating investigation into whether it was a helpful decision.

In the PT's case, an additional functionality is introduced. While the exercise list maintains the same appearance, clicking on the exercise reveals a text box which allows modifications to be made to the instructions. The intuitiveness and ease needed to be ascertained.